SIMATS ENGINEERING
(SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1

Research Paper Review / Literature Survey Report

Topic 30: Smart Forest Fire Detection and Environmental Monitoring System

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Abstract

Forest ecosystems face escalating threats from wildfires that have intensified in frequency, scale, and severity due to climate change, prolonged droughts, and expanding human activity at forest margins. Conventional fire-detection approaches, including manual watch-tower observation and periodic satellite overpasses, typically identify a fire only after it has spread across a substantial area, leaving little time for containment. This literature survey examines the assigned problem statement, the design of a Smart Forest Fire Detection and Environmental Monitoring System capable of continuously sensing environmental parameters such as temperature, humidity, smoke concentration, and wind speed, and of applying artificial intelligence to forecast fire risk, detect ignition events at an early stage, and generate alerts for emergency responders.

Fifteen peer-reviewed and conference research works spanning Internet of Things (IoT) sensor networks, wireless communication architectures, satellite and optical remote sensing, unmanned aerial vehicle (UAV) imaging, and deep-learning-based prediction models were reviewed and compared. The survey summarises the objectives, methodologies, datasets, and key findings of each study, evaluates the approaches against parameters such as detection accuracy, response latency, scalability, and cost, and identifies limitations including high false-alarm rates, limited connectivity in remote terrain, dependence on cloud-based processing, and the absence of unified platforms that fuse ground sensors, drones, and satellite data. Based on the identified research gaps, the report proposes future directions for a multi-layered, AI-driven forest-fire monitoring platform and concludes with a consolidated set of design recommendations relevant to the assigned problem statement.

1. Introduction

1.1 Industry Context

Forest fires cause severe ecological damage, biodiversity loss, and economic impact worldwide, and climate change has increased both the frequency and the intensity of wildfire events, making early detection an urgent priority for governments and conservation agencies. Artificial intelligence, Internet of Things (IoT) sensor networks, satellite imagery, unmanned aerial vehicles, and cloud-based analytics now make continuous, wide-area forest monitoring technically and economically feasible. Public agencies, forestry departments, and disaster-management authorities are increasingly combining these technologies to move from reactive firefighting toward proactive prevention and rapid early response.

1.2 Problem Statement

Conventional forest-fire detection methods rely on manual observation from watch towers or on periodic satellite passes, both of which typically detect a fire only after it has already spread over a significant area. This delay results in extensive environmental damage, loss of biodiversity, and considerably more difficult and costly firefighting operations. A smart forest-fire monitoring platform is therefore required that continuously collects environmental data such as temperature, humidity, smoke concentration, and wind speed through distributed IoT sensors, applies artificial-intelligence models to analyse these parameters, predicts fire risk before ignition where possible, detects fire incidents at the earliest possible stage, generates immediate alerts, and assists emergency-response agencies in controlling the spread of wildfires.

1.3 Objectives

- Monitor forest environmental conditions continuously using distributed sensor networks.
- Detect forest fires at the earliest possible stage to minimise spread.
- Predict fire risk using artificial-intelligence and machine-learning models.
- Integrate drone and satellite monitoring with ground-level sensing.
- Generate timely emergency alerts for forest authorities and firefighting teams.
- Support firefighting operations with real-time situational data.
- Produce environmental analytics reports for long-term forest management.
- Improve forest conservation and biodiversity protection outcomes.

1.4 Expected Outcomes

- Early forest-fire detection with reduced false-alarm rates.
- Reduced environmental damage from faster containment.
- Improved emergency response coordination and resource allocation.
- Better continuous forest health and micro-climate monitoring.
- Enhanced biodiversity protection through habitat preservation.
- Reduced wildfire spread through earlier intervention.
- Data-driven forest management supported by analytics dashboards.
- Long-term, sustainable environmental conservation outcomes.

The remainder of this report presents a literature survey of fifteen research works relevant to the assigned problem statement, a comparative analysis of the surveyed approaches, an identification of research gaps, and recommended future directions and a proposed system architecture.

2. Literature Survey

This section reviews fifteen research works relevant to the assigned problem statement, grouped into four themes: IoT-based sensor and communication approaches, satellite and optical remote-sensing approaches, artificial-intelligence and deep-learning approaches, and UAV/drone-based approaches. Each review summarises the objective, methodology, dataset or tools used, key findings, limitations, and the specific research gap identified for that study.

2.1 IoT-Based Sensor and Communication Approaches

The following eight studies design and, in most cases, prototype ground-level IoT sensor networks for continuous environmental monitoring and fire alerting, differing mainly in their sensing modality, communication backbone, and degree of on-board intelligence.

2.1.1 Cluster-Based Wireless Surveillance for Mountain Forests (FFSS)

Author / Source: Forest-Fire Surveillance System (FFSS) study, South Korea

Objective: Develop a real-time surveillance platform for mountainous forest terrain that overcomes the coverage and connectivity limits of manual patrols.

Methodology and Technology: A wireless sensor network arranged in a cluster-tree topology gathers temperature, humidity, and smoke readings from distributed ground nodes; a service-oriented backend fuses the readings with Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) classifiers to flag likely ignition events.

Dataset / Tools: Field-deployed sensor nodes across a high-risk wildfire zone in South Korean mountains, combined with historical incident records used to train the CNN/LSTM classifiers.

Key Findings: The service-oriented, learning-based pipeline reported detection accuracy above 95 percent and roughly a 40 percent reduction in response time relative to threshold-only detection, and the authors judged the design scalable to additional high-risk zones.

Limitations: The cluster-tree topology depends on reliable multi-hop relay links, which are difficult to sustain in dense canopy or after nodes are damaged by fire or falling debris; the study also stops short of integrating satellite or drone confirmation.

Identified Research Gap: No mechanism is proposed for cross-validating ground-sensor alerts against aerial or satellite imagery, leaving the system exposed to sensor drift and localized false positives.

2.1.2 NodeMCU-Based Early Warning Prototype for Tropical Forests

Author / Source: IoT early-warning prototype, Malaysia (Springer, 2025)

Objective: Provide a low-cost, rapidly deployable early-warning device suited to Malaysia's hot, dry-season forest conditions.

Methodology and Technology: A NodeMCU microcontroller is interfaced with a flame sensor, an MQ2 smoke sensor, and a DHT11 temperature/humidity sensor; readings are pushed to the Blynk mobile application so that forestry staff receive alerts directly on a smartphone, and the electronics are housed in a 3D-printed weatherproof enclosure.

Dataset / Tools: Prototype-level bench and field testing across multiple simulated forest sectors rather than a large historical dataset.

Key Findings: The prototype achieved high detection accuracy and a fast alert response time during testing, and the authors argue that deploying multiple low-cost units across forest sectors gives comprehensive spatial coverage.

Limitations: The evaluation is limited to prototype-scale trials rather than long-term multi-season deployment, and MQ-series gas sensors are known to drift and to respond to non-fire smoke or dust sources, which can raise false-alarm rates over time.

Identified Research Gap: There is no predictive layer: the device reacts only once smoke or heat is already present, so it cannot anticipate fire risk ahead of ignition.

2.1.3 PIR-Flame and GSM-Based Detection and Protection System

Author / Source: IoT-based Forest Fire Detection and Protection System (IEEE, 2023)

Objective: Build an interactive fire detector that both senses ignition and notifies forestry authorities without relying on continuous internet connectivity.

Methodology and Technology: A NodeMCU-driven unit combines a temperature sensor, a smoke sensor, and a passive-infrared (PIR) flame sensor; on detection, a buzzer sounds locally while a Wi-Fi/GSM channel sends a text message or call alert to the responsible authority.

Dataset / Tools: Bench and controlled-environment testing of the sensor-fusion logic; no large public dataset is used.

Key Findings: The authors report that, according to prior surveys, roughly 80 percent of fire-related losses could be prevented with sufficiently early detection, and their combined sensor/alert design was able to trigger both a local and a remote alert promptly upon simulated ignition.

Limitations: GSM-based alerting can be unreliable in areas with weak cellular coverage, and PIR flame sensors have a limited line-of-sight detection range, making the system best suited to perimeter rather than deep-forest monitoring.

Identified Research Gap: The work does not address how alerts from many distributed units would be aggregated, prioritised, or cross-checked at a central command centre during a large-scale incident.

2.1.4 Threshold-Validated Smart Sensor System

Author / Source: IoT-Based Smart System for Fire Detection in Forests (Springer, 2024)

Objective: Detect forest fires early using an inexpensive combination of temperature, moisture, and humidity sensing with threshold-based validation.

Methodology and Technology: Distributed nodes continuously sample temperature, moisture, and humidity; a threshold-comparison algorithm validates whether a reading pattern indicates fire before an alarm is raised at the affected location, and processed data is transmitted to an authorised recipient over the internet.

Dataset / Tools: Simulated environmental readings and controlled threshold-triggering tests rather than a labelled wildfire dataset.

Key Findings: Threshold validation was found to improve alarm reliability compared with single-sensor triggering, reducing spurious alerts caused by transient environmental fluctuations.

Limitations: Fixed thresholds do not adapt to seasonal or regional baseline variation, so a threshold tuned for one forest type may under- or over-trigger in another, and the system offers no forecasting capability.

Identified Research Gap: An adaptive or learning-based thresholding mechanism, capable of adjusting to local climatic baselines, is not explored.

2.1.5 LoRaWAN and ESP32-Based Remote Forest Monitoring

Author / Source: Developing and Implementation of an IoT System for Early Detection of Forest Fires (Springer, 2025)

Objective: Extend continuous IoT-based fire monitoring to remote forest regions lacking conventional network infrastructure.

Methodology and Technology: ESP32 microcontrollers paired with a LoRaWAN long-range communication network transmit sensor readings to a web-based real-time monitoring platform, enabling coverage of remote forested areas beyond typical Wi-Fi range.

Dataset / Tools: Field deployment and prototype validation of the ESP32/LoRaWAN network and its associated monitoring dashboard.

Key Findings: The deployed system achieved a marked reduction in detection-to-response time and demonstrated high accuracy with few false positives or false negatives during field trials.

Limitations: LoRaWAN offers long range but limited bandwidth, restricting the system to lightweight telemetry rather than image or video transmission, and gateway placement still requires line-of-sight planning in hilly terrain.

Identified Research Gap: The study does not integrate imagery-based confirmation, so alerts still rely solely on scalar environmental readings rather than visual verification.

2.1.6 Satellite-Relayed Sensor Network for Wildfire Detection

Author / Source: Sensor-Based Satellite IoT for Early Wildfire Detection (arXiv, 2021)

Objective: Overcome the connectivity gap of terrestrial IoT networks in remote wilderness by relaying sensor data through satellite links.

Methodology and Technology: Distributed ground sensors measuring fire-relevant parameters transmit data via a satellite backhaul rather than terrestrial cellular or Wi-Fi networks, drawing on prior wireless-sensor-network and watchtower-optimisation literature for node placement strategy.

Dataset / Tools: Design study informed by prior wildfire wireless-sensor-network deployments and by watchtower-deployment optimisation research rather than a new field trial.

Key Findings: Satellite backhaul was shown to remove the terrestrial coverage constraint entirely, making continuous monitoring feasible in areas with no cellular infrastructure at all.

Limitations: Satellite communication introduces higher per-message latency and cost than terrestrial links, and narrowband satellite IoT channels constrain the volume of data, such as imagery, that can be transmitted per alert.

Identified Research Gap: Cost-effective hybrid architectures that combine terrestrial mesh coverage with satellite backhaul only where terrestrial connectivity is absent are not fully developed.

2.1.7 FDPA: Detection, Prediction, and Behaviour Analysis System

Author / Source: Alkhatib et al., FDPA IoT System (IET Wireless Sensor Systems, 2024)

Objective: Move beyond detection alone to also predict fire occurrence hours in advance and analyse the direction and speed of fire spread.

Methodology and Technology: A wireless sensor network processes environmental parameters, including temperature, relative humidity, and the Chandler Burning Index, comparing them against ecological thresholds to both anticipate ignition and, once a fire starts, estimate its spread direction and speed.

Dataset / Tools: Experimental deployment and simulation using ecological parameter thresholds calibrated against known fire-behaviour indices such as the Chandler Burning Index.

Key Findings: The FDPA system was able to anticipate natural ignition several hours ahead of occurrence in testing and to provide real-time estimates of fire spread direction and speed to support emergency responders.

Limitations: Predictive accuracy is tied closely to the accuracy of the Chandler Burning Index for the specific vegetation type and region, which may not generalise well across highly heterogeneous forest ecosystems.

Identified Research Gap: Integration with satellite-derived vegetation-dryness indices to improve the generalisability of the predictive model across different forest biomes is left for future work.

2.1.8 Real-World Efficacy Assessment Using SEA-HAZEMON Data

Author / Source: Belcher, Esteva, Lam, Ramadhani, Rayhan, Xu, and Tuncer (Imperial College London / arXiv, 2024)

Objective: Empirically assess how well distributed low-cost IoT sensor nodes perform at fire and haze detection under real, rather than simulated, field conditions.

Methodology and Technology: The study analyses real sensor data collected by the SEA-HAZEMON (Low-Cost IoT Sensor of Haze Air Quality Disasters) platform deployed across Thailand and Southeast Asia, using it as a case study to derive practical design guidelines for a 6G-enabled IoT fire-detection mechanism.

Dataset / Tools: Real-world haze and air-quality sensor readings from the SEA-HAZEMON deployment across Thailand and Southeast Asian sites.

Key Findings: The case study exposed practical challenges in achieving reliable detection with distributed low-cost nodes under field conditions and proposed a preliminary set of guidelines for designing next-generation, 6G-enabled IoT fire-detection networks.

Limitations: The authors explicitly frame the work as preliminary; sensor calibration drift, node maintenance, and inconsistent connectivity were all observed to affect real-world reliability more than laboratory results suggested.

Identified Research Gap: A validated, production-ready 6G IoT architecture for wildfire detection has not yet been demonstrated; the paper identifies the need but does not deliver the full system.

2.2 Satellite, Optical, and Remote-Sensing Approaches

The next two studies examine detection and prediction using optical and satellite remote sensing, which trade the fine spatial granularity of ground sensors for wide-area, infrastructure-free coverage.

2.2.1 Systematic Review of Optical Remote-Sensing Fire Detection

Author / Source: Barmpoutis, Papaioannou, Dimitropoulos, and Grammalidis (MDPI Sensors, 2020)

Objective: Survey optical remote-sensing technologies and flame/smoke detection algorithms used across terrestrial, airborne, and spaceborne early-warning systems.

Methodology and Technology: A structured literature review classifies existing fire-detection systems into terrestrial camera-based, airborne, and spaceborne categories, and compares the flame- and smoke-detection algorithms associated with each.

Dataset / Tools: Secondary analysis of previously published detection systems and their reported test imagery rather than a new primary dataset.

Key Findings: Terrestrial camera systems were found to offer the fastest detection but the smallest coverage area, spaceborne systems the largest coverage but the coarsest spatial and temporal resolution, and airborne systems a middle ground, with hybrid combinations recommended for balanced performance.

Limitations: Because the paper is a review rather than an original system, it does not provide new experimental accuracy figures, and the surveyed systems it discusses each have their own individual weaknesses in smoke-versus-cloud discrimination.

Identified Research Gap: A standardised benchmark for comparing terrestrial, airborne, and spaceborne detection algorithms on the same fire events is identified as missing from the field.

2.2.2 Machine-Learning Pipeline for Satellite-Based Fire Detection and Prediction

Author / Source: Deep Learning Approaches for Forest Fires Detection and Prediction using Satellite Images (Procedia Computer Science, 2024)

Objective: Combine classical machine learning and deep learning within a single pipeline to both detect active fires and classify fire risk from satellite data.

Methodology and Technology: Satellite imagery is calibrated, filtered, and normalised before an AI layer applies Random Forest and logistic-regression models alongside Multilayer Perceptron (MLP) and LSTM deep-learning models to classify fire risk levels.

Dataset / Tools: Multispectral satellite imagery processed through a calibration and normalisation pipeline prior to classification.

Key Findings: The combined classical/deep-learning pipeline improved fire-risk classification consistency over using either family of models alone, and the paper situates its contribution alongside related CNN-based active-fire-detection work using Himawari-8 and geostationary satellite data.

Limitations: Satellite-based classification remains constrained by revisit time and cloud cover, meaning that fast-moving fires can begin and spread between successive satellite passes.

Identified Research Gap: Fusion of the satellite-derived risk classification with ground-based real-time sensor confirmation, to compensate for satellite revisit-time gaps, is not implemented.

2.3 Artificial Intelligence and Deep-Learning Approaches

These three studies focus specifically on the machine-learning and deep-learning modelling layer used to convert raw sensor or imagery data into fire-risk predictions and spread forecasts.

2.3.1 Deep-Learning Wildfire Susceptibility Mapping Across Two Satellite Datasets

Author / Source: Wildfire Susceptibility Mapping Using Deep Learning Algorithms in Two Satellite Imagery Datasets (MDPI Forests, 2023)

Objective: Compare deep-learning architectures for producing wildfire susceptibility maps from two independent satellite sources.

Methodology and Technology: LSTM and Recurrent Neural Network (RNN) models are trained on twelve wildfire-influencing factors using MODIS and Landsat-8 imagery covering wildfire events from 2021, and their performance is benchmarked against CNN, Deep Neural Network (DNN), and LSTNet baselines.

Dataset / Tools: 599 wildfire locations paired with Landsat-8 imagery and 232 sites paired with MODIS imagery, each annotated with twelve influencing factors.

Key Findings: The LSTM model achieved approximately 96 percent accuracy, comparable to more complex attention-based models, while the plain CNN and DNN baselines trailed at roughly 88 percent and 75 percent respectively, indicating that temporal modelling of satellite time series materially improves susceptibility mapping.

Limitations: The comparatively small number of labelled wildfire sites (599 and 232) limits the statistical robustness of the reported accuracy figures, and the models are evaluated on historical events rather than in a live operational setting.

Identified Research Gap: Real-time operational deployment and validation of the susceptibility-mapping model against newly occurring fires, rather than retrospective historical events, is not demonstrated.

2.3.2 Comprehensive Survey of the Machine-Learning Pipeline for Wildfire Risk Assessment

Author / Source: A Comprehensive Survey of the Machine Learning Pipeline for Wildfire Risk Prediction and Assessment (ScienceDirect, 2025)

Objective: Consolidate the end-to-end machine-learning pipeline used across the wildfire-risk-prediction literature, from data sources to operational decision-support systems.

Methodology and Technology: The survey reviews and organises CNN-, RNN-, and LSTM-based susceptibility-mapping studies, unsupervised clustering methods for regional risk grouping, and reinforcement-learning approaches used for evacuation and resource-allocation decision-making, including an examination of the U.S. Forest Service's FireCast platform and IBM's wildfire-management tools.

Dataset / Tools: Secondary review of multiple published datasets and operational platforms rather than a single primary dataset.

Key Findings: The survey concludes that deep-learning models generally outperform traditional machine-learning techniques for susceptibility mapping, and that few existing systems combine detection, prediction, and operational decision-support (such as evacuation planning) within a single unified pipeline.

Limitations: As a survey, it does not propose or validate a new unified architecture, and it notes that many of the reviewed operational platforms are proprietary, limiting reproducibility.

Identified Research Gap: The survey explicitly highlights the absence of an openly documented, end-to-end pipeline that spans data ingestion, risk prediction, detection, and automated response coordination.

2.3.3 Systematic Review of Deep Learning for Wildfire Spread Prediction

Author / Source: Machine Learning and Deep Learning for Wildfire Spread Prediction: A Review (MDPI, 2024)

Objective: Review the datasets, model families, and reported improvements for predicting how an active wildfire will spread once ignited.

Methodology and Technology: The review compares classical empirical spread models, machine-learning approaches such as support vector machines and ensemble methods, and deep-learning approaches such as CNNs and convolutional recurrent networks (CRNs), summarising which architectures best capture the spatial and temporal complexity of fire spread.

Dataset / Tools: Secondary review of multiple public wildfire-spread datasets used across the surveyed studies.

Key Findings: CNN-based models were found to be particularly effective for spatial pattern extraction from satellite imagery, while CRN-based models performed better when both spatial and temporal dynamics needed to be captured jointly, generally outperforming classical empirical spread models.

Limitations: The review notes that classical spread models remain in operational use in many agencies because deep-learning models require substantial computational infrastructure and labelled training data that are not always available at the local level.

Identified Research Gap: Lightweight, edge-deployable spread-prediction models suitable for real-time use directly at forest monitoring stations, rather than in a centralised cloud, are identified as underexplored.

2.4 UAV / Drone-Based Approaches

The final two studies extend detection into the air, using unmanned aerial vehicles either for high-resolution segmentation of fire-affected regions or for autonomous, AI-guided response following a ground-sensor alert.

2.4.1 UAV-Based Fire Segmentation Using DeepFire S3GA-Net

Author / Source: Automated Forest Fire Detection in Ecological Monitoring Using Enhanced Deep Learning Networks (Scientific Reports, 2025)

Objective: Improve the accuracy of forest-fire region segmentation from unmanned aerial vehicle (UAV) imagery under variable fire size, smoke density, terrain, and lighting conditions.

Methodology and Technology: DeepFire S3GA-Net, a novel deep-learning segmentation framework, is applied to UAV-captured aerial imagery and benchmarked against prior approaches such as MS-FRCNN, SmokeyNet, and Fire-Net, which respectively use region-based CNNs, spatiotemporal static-camera imagery, and multiscale Himawari-8 satellite features.

Dataset / Tools: UAV-collected aerial imagery together with comparative benchmarks drawn from published datasets such as the Fire Ignition Library (FIgLib) used to train SmokeyNet.

Key Findings: DeepFire S3GA-Net outperformed the compared baselines in segmentation accuracy under heterogeneous conditions, and the authors note that prior models such as SmokeyNet had already reached accuracy comparable to human annotators on static-camera imagery, showing the broader field's rapid progress.

Limitations: UAV-based imaging requires either scheduled or on-demand flights, which limits it to intermittent rather than truly continuous coverage compared with fixed ground sensors, and battery endurance constrains the area a single drone can cover per mission.

Identified Research Gap: Autonomous, trigger-based UAV dispatch that launches a drone automatically in response to a ground-sensor alert, rather than on a fixed schedule, is not implemented in this work.

2.4.2 AI-Powered Drones Coupled with LoRa IoT Ground Sensors

Author / Source: Towards Early Forest Fire Detection and Prevention Using AI-Powered Drones and the IoT (ScienceDirect, 2024)

Objective: Combine ground-based IoT sensing with autonomous drone dispatch so that a suspected fire can be visually confirmed and, where feasible, acted upon before it spreads.

Methodology and Technology: LoRa-based IoT ground nodes, which enter a deep-sleep mode between transmissions to conserve power, cover a wide forest area and relay readings to gateways; on a suspected detection, an AI-based pipeline directs a drone to the location to visually track the fire centre, warn any people present through a two-way audio channel, and alert authorities, with several drone-gateway connectivity architectures evaluated.

Dataset / Tools: Simulation and prototype testing of static-gateway, mobile drone-borne gateway, and cloud-connected sensing-node architectures.

Key Findings: The AI-based tracking pipeline was able to accurately detect and follow the centre of a fire once a drone reached the site, and the study compared multiple connectivity architectures to identify trade-offs between coverage, latency, and power consumption.

Limitations: Drone flight time, payload capacity (for any extinguishing equipment), and adverse weather conditions such as high wind or heavy smoke restrict the operational envelope of the aerial response component.

Identified Research Gap: Long-term field validation across multiple fire seasons and forest types, beyond the reported simulation and prototype trials, is not yet available.

3. Comparative Literature Survey Table

Table 1 consolidates the author/year, research problem, methodology, dataset or tools, key findings, and the principal limitation or research gap for each of the fifteen reviewed studies, in line with the comparative-analysis requirement of this assessment.

Table 1. Comparative summary of reviewed literature

Ref.	Author / Year	Research Problem	Methodology	Dataset / Tools	Key Findings	Limitation / Research Gap
[1]	Forest-Fire Surveillance System (FFSS) study, South Korea	Develop a real-time surveillance platform for mountainous forest terrain that overcomes the coverage and connectivity limits of manual patrols.	Cluster-tree WSN + CNN/LSTM classifier	Field nodes, historical incident records	>95% accuracy, 40% faster response	No aerial/satellite cross-validation
[2]	IoT early-warning prototype, Malaysia (Springer, 2025)	Provide a low-cost, rapidly deployable early-warning device suited to Malaysia's hot, dry-season forest conditions.	NodeMCU + flame/MQ2/DHT11, Blynk alerts	Prototype bench/field tests	High accuracy, fast alerts	No predictive capability
[3]	IoT-based Forest Fire	Build an interactive fire	NodeMCU + PIR flame + GSM/Wi-Fi alert	Controlled bench tests	Local + remote alerting	No multi-node alert aggregation

Ref.	Author / Year	Research Problem	Methodology	Dataset / Tools	Key Findings	Limitation / Research Gap
	Detection and Protection System (IEEE, 2023)	detector that both senses ignition and notifies forestry authorities without relying on continuous internet connectivity.				
[4]	IoT-Based Smart System for Fire Detection in Forests (Springer, 2024)	Detect forest fires early using an inexpensive combination of temperature, moisture, and humidity sensing with threshold-based validation.	Threshold-validated temp/moisture/humidity sensing	Simulated environmental readings	Fewer spurious alarms	Fixed, non-adaptive thresholds
[5]	Developing and Implementation of an IoT System for Early Detection of Forest Fires	Extend continuous IoT-based fire monitoring to remote forest regions lacking conventional	ESP32 + LoRaWAN + web dashboard	Field deployment, prototype trial	Reduced response time, high accuracy	No imagery-based confirmation

Ref.	Author / Year	Research Problem	Methodology	Dataset / Tools	Key Findings	Limitation / Research Gap
	(Springer, 2025)	network infrastructure.				
[6]	Sensor-Based Satellite IoT for Early Wildfire Detection (arXiv, 2021)	Overcome the connectivity gap of terrestrial IoT networks in remote wilderness by relaying sensor data through satellite links.	Satellite-relayed ground sensors	Design study, prior WSN literature	Removes terrestrial coverage limit	High latency/cost, narrow bandwidth
[7]	Alkhatib et al., FDPA IoT System (IET Wireless Sensor Systems, 2024)	Move beyond detection alone to also predict fire occurrence hours in advance and analyse the direction and speed of fire spread.	WSN + Chandler Burning Index thresholds	Ecological parameter thresholds	Predicts ignition hours ahead	Index accuracy is region/vegetation specific
[8]	Belcher, Esteva, Lam, Ramadhani, Rayhan, Xu, and Tuncer (Imperial College	Empirically assess how well distributed low-cost IoT sensor nodes perform at fire and haze	Field analysis of low-cost IoT nodes (SEA-HAZEMON)	Real haze/air-quality sensor data	Practical 6G IoT design guidelines	Preliminary; no production system

Ref.	Author / Year	Research Problem	Methodology	Dataset / Tools	Key Findings	Limitation / Research Gap
	London / arXiv, 2024)	detection under real, rather than simulated, field conditions.				
[9]	Barmpoutis, Papaioannou, Dimitropoulos, and Grammalidis (MDPI Sensors, 2020)	Survey optical remote-sensing technologies and flame/smoke detection algorithms used across terrestrial, airborne, and spaceborne early-warning systems.	Systematic review: terrestrial/airborne/spaceborne optical	Secondary review of prior systems	Hybrid combinations recommended	No standard cross-system benchmark
[10]	Deep Learning Approaches for Forest Fires Detection and Prediction using Satellite Images (Procedia Computer Science, 2024)	Combine classical machine learning and deep learning within a single pipeline to both detect active fires and classify fire risk from satellite data.	RF, logistic regression, MLP, LSTM on satellite imagery	Calibrated multispectral imagery	Improved risk-classification consistency	Limited by satellite revisit time/cloud cover

Ref.	Author / Year	Research Problem	Methodology	Dataset / Tools	Key Findings	Limitation / Research Gap
[11]	Wildfire Susceptibility Mapping Using Deep Learning Algorithms in Two Satellite Imagery Datasets (MDPI Forests, 2023)	Compare deep-learning architectures for producing wildfire susceptibility maps from two independent satellite sources.	LSTM/RNN vs CNN/DNN/LSTNet on MODIS & Landsat-8	599 & 232 labelled wildfire sites	LSTM ~96% accuracy	Small labelled sample; retrospective only
[12]	A Comprehensive Survey of the Machine Learning Pipeline for Wildfire Risk Prediction and Assessment (ScienceDirect, 2025)	Consolidate the end-to-end machine-learning pipeline used across the wildfire-risk-prediction literature, from data sources to operational decision-support systems.	Survey of CNN/RNN/LSTM, clustering, reinforcement learning	Secondary review of platforms (FireCast, IBM)	DL > traditional ML for mapping	No unified open end-to-end pipeline
[13]	Machine Learning and Deep Learning for Wildfire Spread Prediction: A	Review the datasets, model families, and reported improvements	Review of empirical, ML (SVM), and DL (CNN/CRN) spread models	Secondary review of public datasets	CRNs best for spatio-temporal spread	Lacks lightweight edge-deployable models

Ref.	Author / Year	Research Problem	Methodology	Dataset / Tools	Key Findings	Limitation / Research Gap
	Review (MDPI, 2024)	for predicting how an active wildfire will spread once ignited.				
[14]	Automated Forest Fire Detection in Ecological Monitoring Using Enhanced Deep Learning Networks (Scientific Reports, 2025)	Improve the accuracy of forest-fire region segmentation from unmanned aerial vehicle (UAV) imagery under variable fire size, smoke density, terrain, and lighting conditions.	DeepFire S3GA-Net UAV segmentation	UAV imagery, FIgLib benchmark	Outperforms MS-FRCNN, SmokeyNet, Fire-Net	No autonomous trigger-based dispatch
[15]	Towards Early Forest Fire Detection and Prevention Using AI-Powered Drones and the IoT (ScienceDirect, 2024)	Combine ground-based IoT sensing with autonomous drone dispatch so that a suspected fire can be visually confirmed and, where	LoRa IoT nodes + AI-guided drone dispatch	Simulated/prototype connectivity architectures	Accurate fire-centre tracking by drone	Limited flight time; no multi-season validation

Ref.	Author / Year	Research Problem	Methodology	Dataset / Tools	Key Findings	Limitation / Research Gap
		feasible, acted upon before it spreads.				

4. Critical Analysis and Research Gaps

4.1 Fragmentation Across Sensing Modalities

The reviewed literature clusters into largely independent research threads: ground-based IoT sensor networks (studies 1-8), satellite and optical remote sensing (studies 9-10), predictive AI/deep-learning models (studies 11-13), and UAV-based visual confirmation (studies 14-15). Very few of the surveyed works attempt to fuse more than two of these modalities within a single deployed system. Ground sensors offer fast, localised detection but are vulnerable to false positives from dust, insects, or agricultural burning; satellite systems offer wide coverage but suffer from revisit-time gaps and cloud occlusion; UAVs offer high-resolution visual confirmation but only intermittent coverage limited by flight endurance. The assigned problem statement explicitly calls for a platform that integrates IoT sensing, AI prediction, and drone/satellite monitoring together, yet no single reviewed study delivers all three in an operationally validated form.

4.2 High False-Alarm Rates and Sensor Reliability

Several IoT-based studies (notably studies 2, 3, and 4) rely on gas and PIR sensors that are known to drift over time or to respond to non-fire stimuli such as vehicle exhaust, dust, or sunlight glare. Threshold-based validation, as used in study 4, reduces but does not eliminate this problem, since fixed thresholds cannot adapt to seasonal humidity swings or regional vegetation differences. None of the reviewed IoT studies incorporate a secondary, independent confirmation step, such as a quick image capture or a satellite cross-check, before escalating an alert to human responders.

4.3 Connectivity and Coverage Constraints in Remote Terrain

Cellular and Wi-Fi based alerting, used in studies 3 and 4, is unreliable in the dense, hilly, or remote terrain typical of large forest reserves. LoRaWAN (study 5) and satellite backhaul (study 6) both address this gap but introduce their own trade-offs: LoRaWAN's narrow bandwidth precludes transmitting confirmatory imagery, while satellite links add latency and cost that may be unsuitable for time-critical alerts. The survey identifies a clear need for adaptive, hybrid communication architectures that automatically select the lowest-latency available channel (Wi-Fi, cellular, LoRa, or satellite) depending on what infrastructure exists at a given monitoring point.

4.4 Predictive Modelling Limited by Data Scale and Locality

The deep-learning studies reviewed in Section 2.3 report strong accuracy figures, with LSTM-based susceptibility mapping (study 11) reaching approximately 96 percent, but these results are typically obtained on retrospective datasets of a few hundred labelled wildfire sites for a single region. Model generalisation to other forest types, climates, or fire regimes is rarely tested, and the comprehensive survey in study 12 explicitly notes the absence of a publicly documented, end-to-end pipeline spanning data ingestion through to automated response. This represents a significant research gap for any system intended to operate reliably across diverse forest ecosystems, as implied by the assigned problem statement's broad scope.

4.5 Limited Autonomy in Aerial Response

Both UAV-based studies (14 and 15) demonstrate strong visual segmentation and tracking performance once a drone reaches a fire site, but neither demonstrates a fully autonomous pipeline in which a ground-sensor or satellite alert automatically triggers drone dispatch, flight planning, and on-site confirmation without human intervention. Battery endurance and adverse-weather operation also remain open engineering constraints that were not resolved in the reviewed literature.

4.6 Absence of Integrated Analytics and Conservation Reporting

None of the fifteen reviewed studies explicitly addresses the generation of longer-term environmental analytics reports or biodiversity-impact assessments, both of which are called for in the assigned problem statement's objectives. Existing work is almost entirely oriented toward the detection-and-alert cycle, leaving the post-incident and long-term forest-management analytics layer as an open area for future research and system design.

5. Proposed Directions and Future Work

Based on the research gaps identified in Section 4, the following future directions are proposed as practical and directly relevant extensions to the assigned problem statement.

5.1 Multi-Modal Sensor Fusion

Future systems should fuse ground-based IoT readings (temperature, humidity, smoke, wind speed) with satellite-derived vegetation-dryness indices and, where available, periodic UAV imagery, using a weighted confidence-scoring model rather than relying on any single modality to trigger an alert. This directly addresses the fragmentation gap identified in Section 4.1.

5.2 Adaptive, Self-Calibrating Thresholds

Rather than fixed sensor thresholds, an online learning component should continuously recalibrate baseline environmental values for each monitoring zone using recent historical readings, reducing the false-alarm rates observed in several of the reviewed IoT studies.

5.3 Hybrid Adaptive Communication Layer

A communication middleware that automatically selects among Wi-Fi, cellular, LoRaWAN, and satellite backhaul based on real-time link availability and message priority would resolve the connectivity trade-offs identified across studies 3, 5, and 6, ensuring that critical fire alerts are never lost to a single point of network failure.

5.4 Cross-Region Generalisable AI Models

Predictive models should be trained and validated across multiple forest biomes and fire seasons, using transfer-learning techniques to adapt a base model quickly to a new region with only a small amount of local labelled data, addressing the generalisation gap identified in study 12.

5.5 Autonomous, Trigger-Based Drone Dispatch

An automated dispatch controller should be developed that, upon receiving a sufficiently confident ground or satellite alert, autonomously plans and executes a UAV flight to the suspected location for visual confirmation, closing the human-in-the-loop gap identified in studies 14 and 15.

5.6 Integrated Environmental Analytics and Reporting

A dedicated analytics and reporting module should aggregate historical sensor, satellite, and drone data into dashboards for forest managers, tracking biodiversity indicators, vegetation recovery after a fire, and long-term climate trends, addressing the reporting gap identified in Section 4.6.

5.7 Emergency-Response Coordination Interface

A responder-facing interface should translate raw AI outputs into actionable guidance, including estimated fire spread direction and speed (building on the approach demonstrated in study 7) and recommended resource-allocation priorities, to directly support firefighting operations as called for in the assigned problem statement.

6. Proposed System Architecture

Drawing on the strengths and addressing the gaps identified in the reviewed literature, a four-layer architecture is proposed for the Smart Forest Fire Detection and Environmental Monitoring System.

Layer 1: Sensing Layer

Distributed IoT nodes (temperature, humidity, smoke, wind speed sensors) similar to those in studies 1-8 are deployed across the forest, supplemented by periodic satellite passes (as in studies 9-10) and scheduled or on-demand UAV overflights (as in studies 14-15).

Layer 2: Communication Layer

A hybrid communication middleware routes sensor data through the best available channel among Wi-Fi, cellular, LoRaWAN, and satellite backhaul, following the design direction proposed in Section 5.3.

Layer 3: Intelligence Layer

A cloud- or edge-hosted AI pipeline combines CNN/LSTM-based fire-risk prediction (as in studies 11-13) with sensor-fusion confidence scoring (Section 5.1) to decide whether an alert should be raised and, if so, at what priority level.

Layer 4: Response and Analytics Layer

Confirmed alerts trigger autonomous drone dispatch for visual verification (Section 5.5) and are simultaneously routed to an emergency-response coordination interface (Section 5.7); all incident and sensor data feed into a long-term analytics and conservation-reporting dashboard (Section 5.6).

7. Conclusion

This literature survey reviewed fifteen research works addressing the assigned problem statement on smart forest-fire detection and environmental monitoring, spanning IoT sensor networks, satellite and optical remote sensing, artificial-intelligence-based prediction, and UAV-based aerial confirmation. Individually, the surveyed approaches demonstrate strong results within their own scope: IoT sensor networks achieve fast, low-cost local detection; satellite and optical systems provide wide-area coverage; deep-learning models achieve high predictive accuracy on historical data; and UAV-based segmentation delivers precise visual confirmation. However, the comparative analysis and critical review presented in Sections 3 and 4 show that no single reviewed system fully integrates these modalities, adapts reliably across regions and seasons, or closes the loop from prediction through autonomous aerial confirmation to long-term conservation analytics.

The future directions and proposed four-layer architecture presented in Sections 5 and 6 outline a practical path toward a more complete Smart Forest Fire Detection and Environmental Monitoring System that directly satisfies the objectives set out in the assigned problem statement: continuous environmental monitoring, early fire detection, AI-based risk prediction, drone and satellite integration, emergency alerting, firefighting support, environmental analytics, and improved forest conservation outcomes.

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